

IBM ECM System Monitor – Using authority signed certificate

ESM Server Security – Using an authority signed certificate on the
ESM server

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Introduction

Overview

This guide describes in detail how the shipped certificate for the ESM Server console can be exchanged with an authority signed certificate. For others (e.g. self-signed certificates) this can be used as reference, just adopt the process.

Disclaimer

The content of this document is based on ESM Version 5.7.0.0-000. The descriptions and guidelines in this document are for informational purposes only. Up-to-dateness, content completeness, appropriateness and validity for all possible scenarios cannot be guaranteed. All information is provided on an as-is basis. The author is not liable for any errors or omissions in this document or any losses, injuries and damages arising from its use.

If you are planning to set up or configure ESM or to adjust an existing installation, it is absolutely necessary to take into account current security whitepapers, release notes and announcements from the official IBM ECM System Monitor product documentation website.

Purpose of this guide

By default, the ESM Server ships a certificate for the ESM console which is a self-signed certificate of the build server of the software. Customers often have the requirement to exchange the certificate with a certificate valid for their environment.

Requirements

- Server certificate and private key must be requested (ideally as pfx keystore file) – see additional information in first step
- ESM Server Version 5.7.0.0-000 or later (already installed)
- Download and install keystore explorer tool for your OS (this “How To” is on Windows)
<https://keystore-explorer.org/downloads.html>

Step by Step setup

1. Request correct certificate and private key

For requesting the certificate and the private key from your authority some information is relevant.

If you plan to access the ESM Server console with different addresses, such as full or short name or even an alias, all these names must be included in the so called Subject Alternative Name. Provide these names to your authority. In this example I will use the following:

esm-prod-server (alias)

esm-prod-server.de.cenit-group.com (alias2)

smccos8esmdb2 (short name)

smccos8esmdb2.de.cenit-group.com (full name)

Request the certificate and the private key including password – the best is if you request a pfx keystore file that includes both, the certificate and the private key.

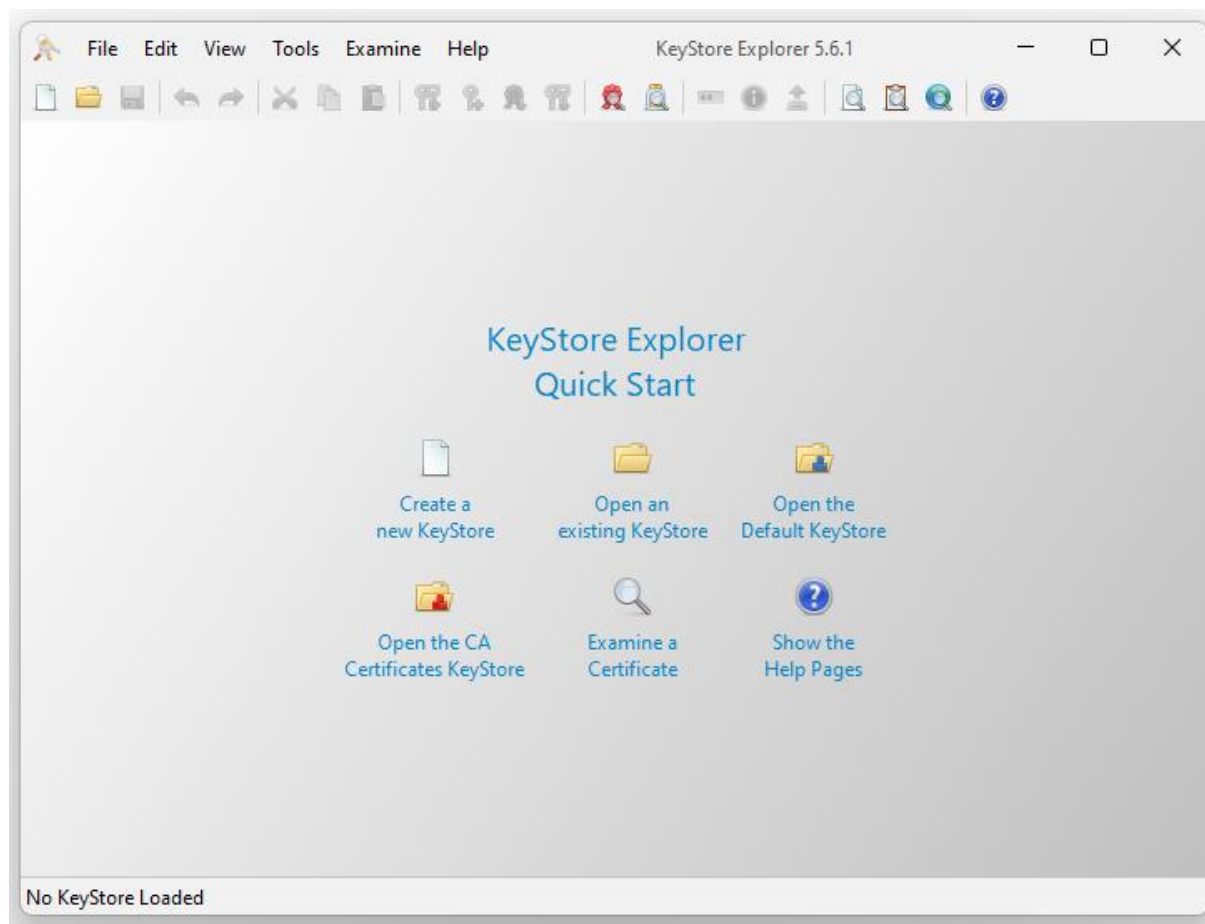
2. Export Private Key and Certificate from pfx

In this part we will be using the keystore explorer tool to export the relevant information from the provided pfx file.

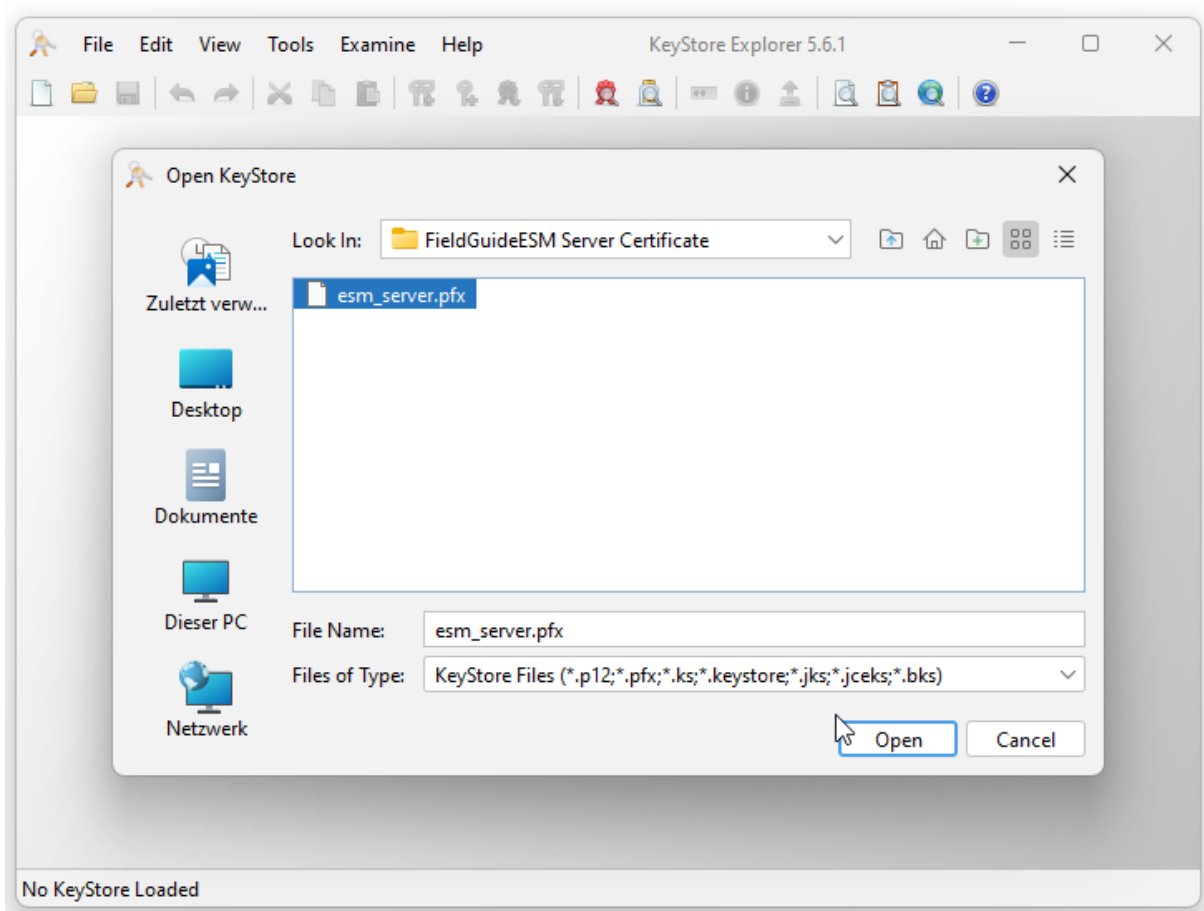
If not already done, please download the tool now and install it.

This description is based on Windows – your screens might look different on other OS types.

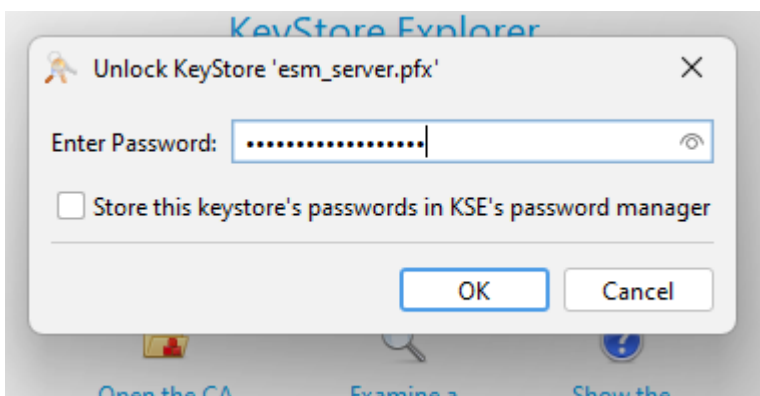
Open the tool:



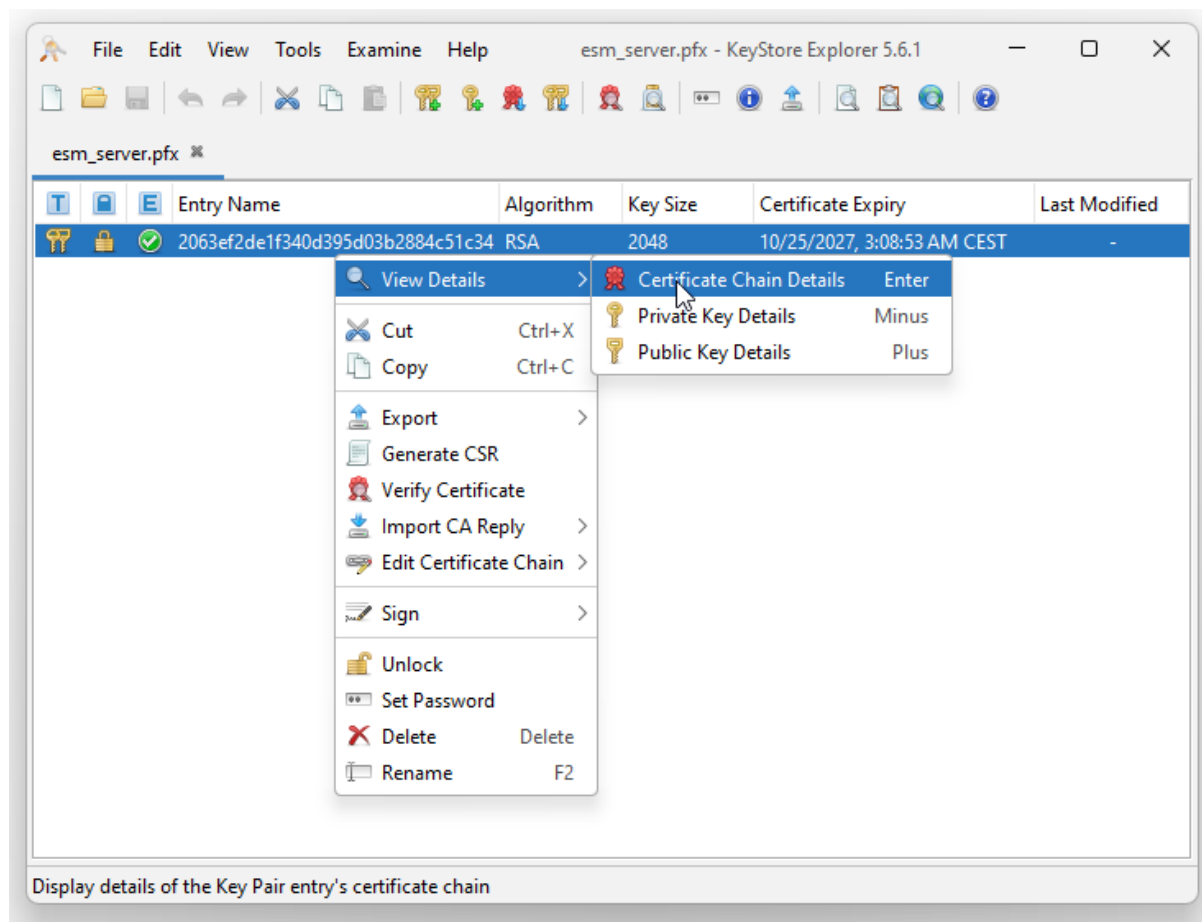
Click on “Open an existing Keystore” and select your pfx keystore file – Click on Open:

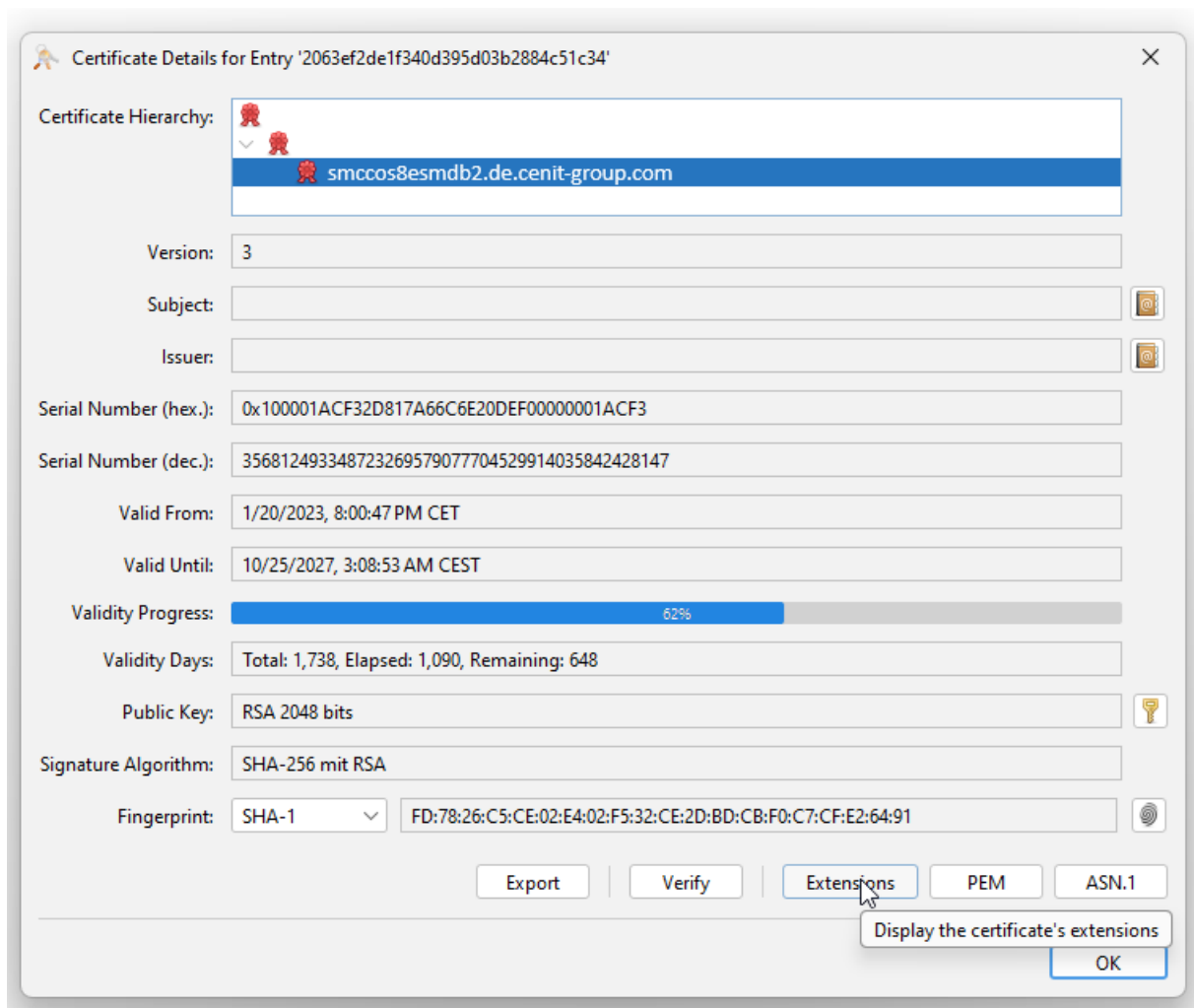


If required, please also unlock the KeyStore with the password that is shipped with the pfx file:



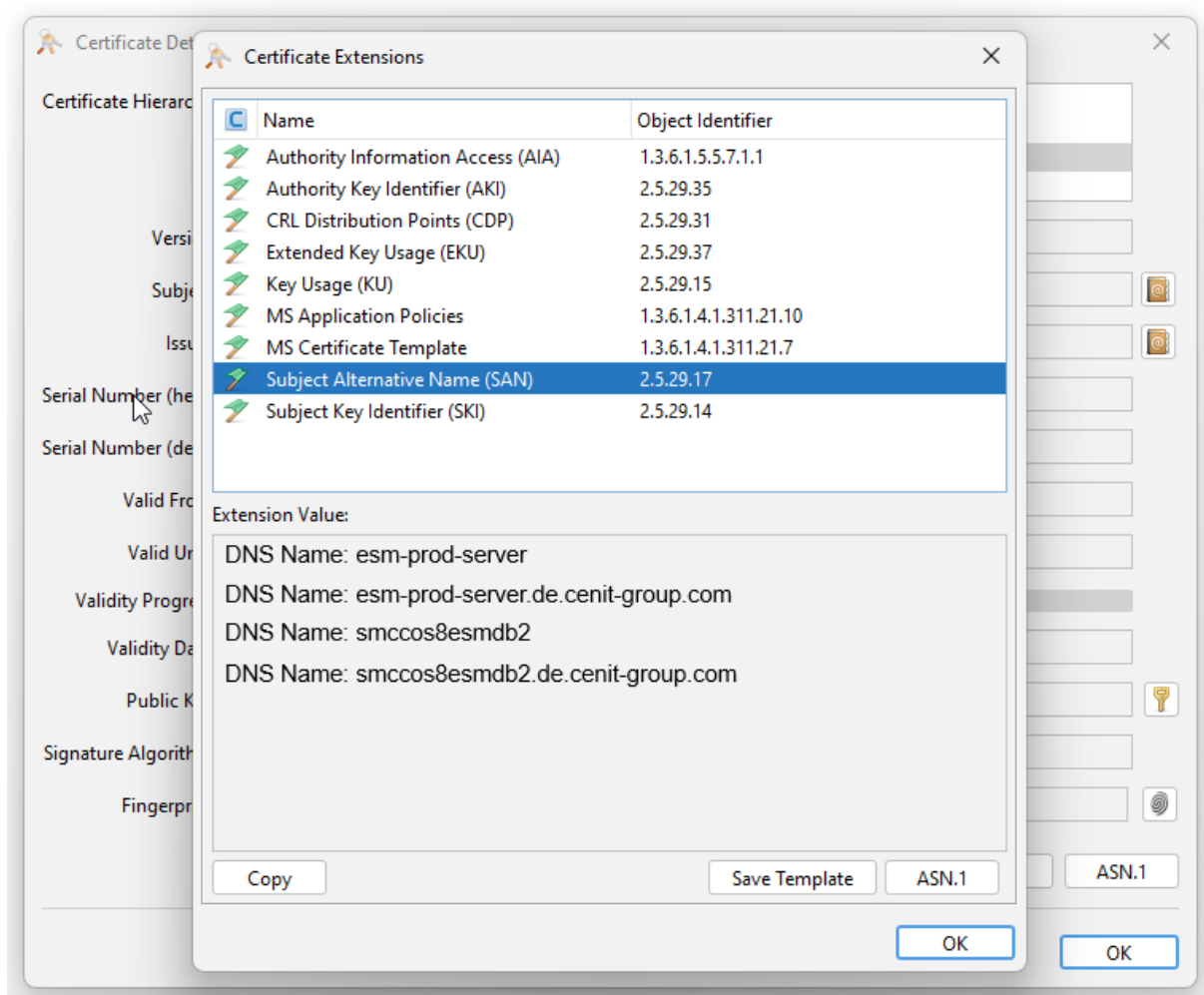
You should now see a certificate in the tool – right click on it and see Certificate Chain Details:





(Note: I removed some information from this screenshot)

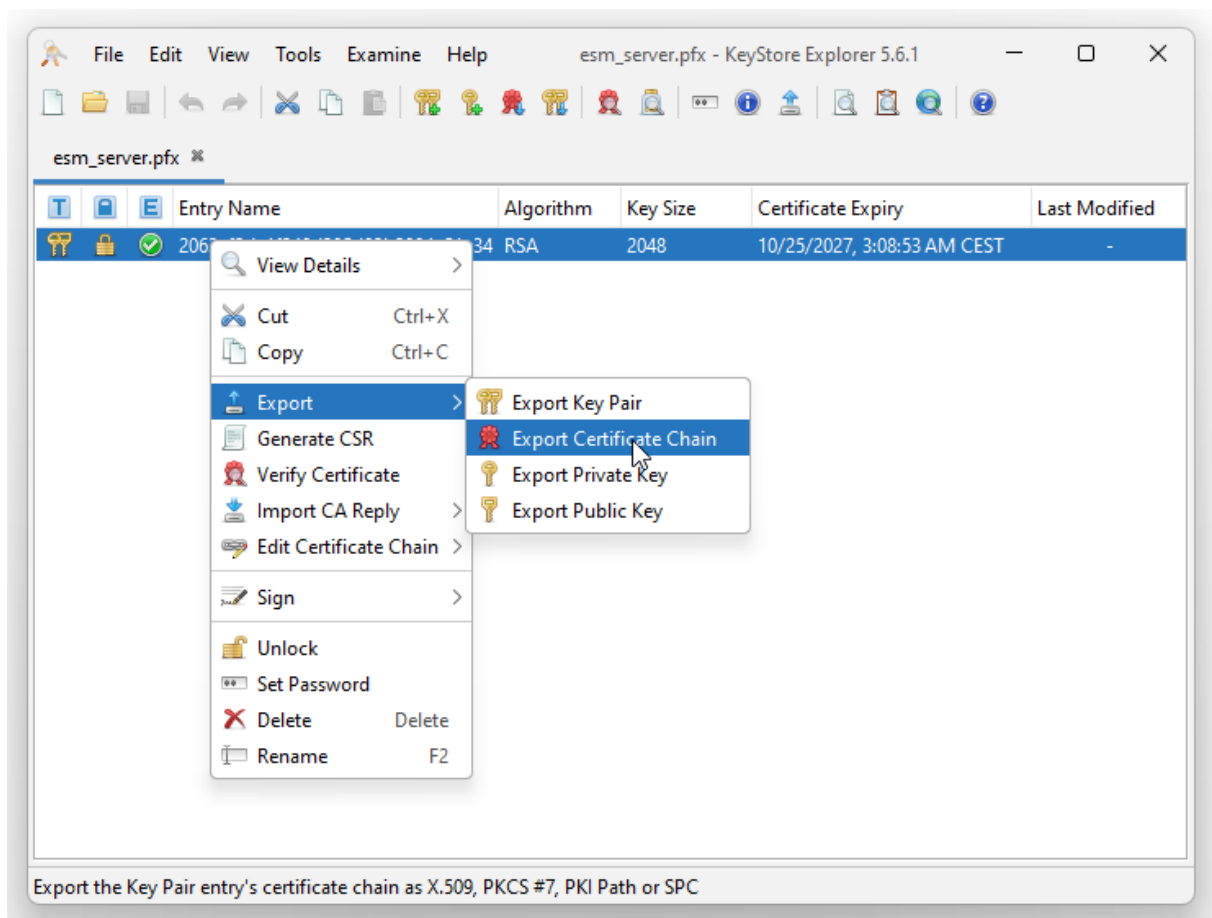
Click on Extensions and select Subject Alternative Name (SAN):



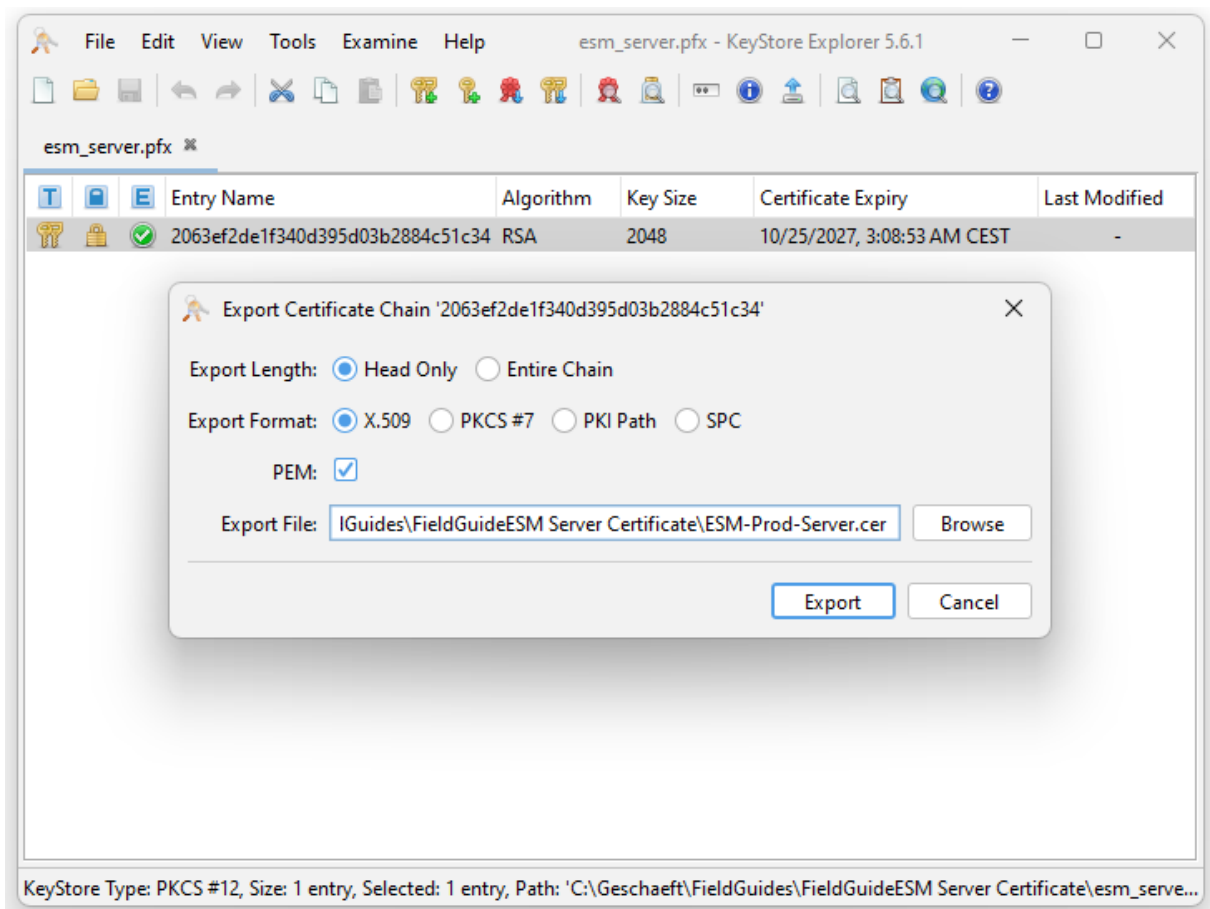
All your requested Subject Alternative Names should be shown here.

Click OK twice to close certificate details.

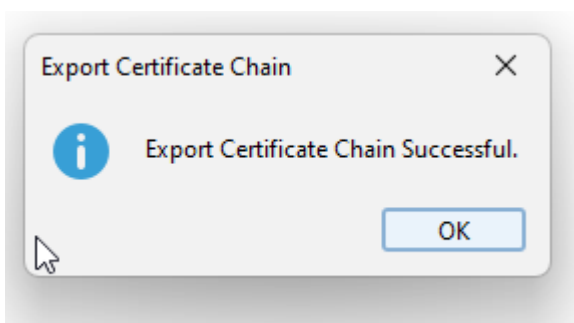
Right click on the certificate again and select Export -> Export Certificate Chain:



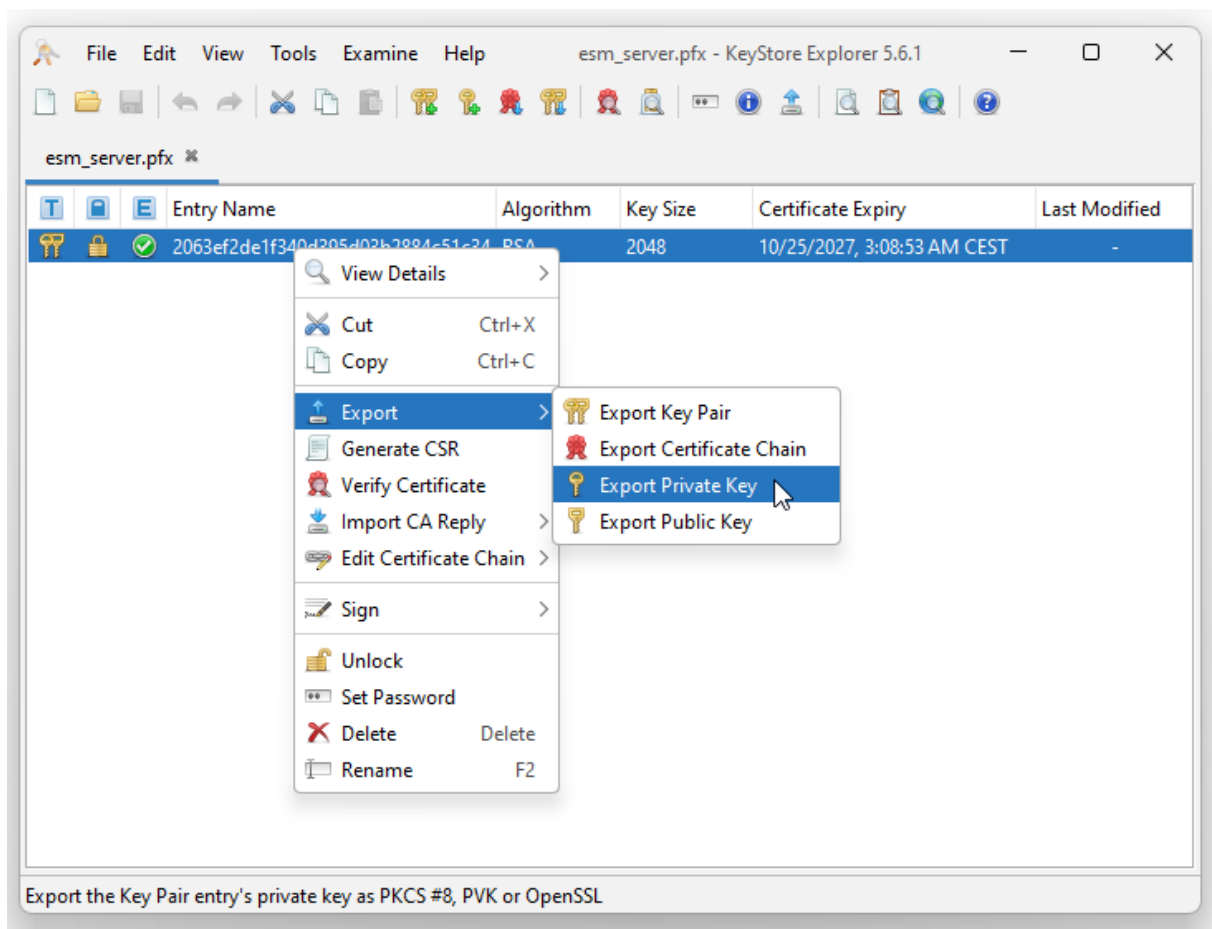
Keep the settings “Head Only”, “X.509” and “PEM” checked. You might want to rename the file to something else – e.g. ESM-Prod-Server.cer and export the file to your destination folder:



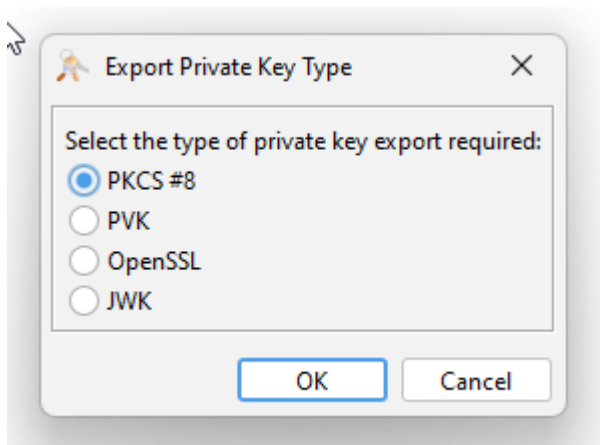
Click OK



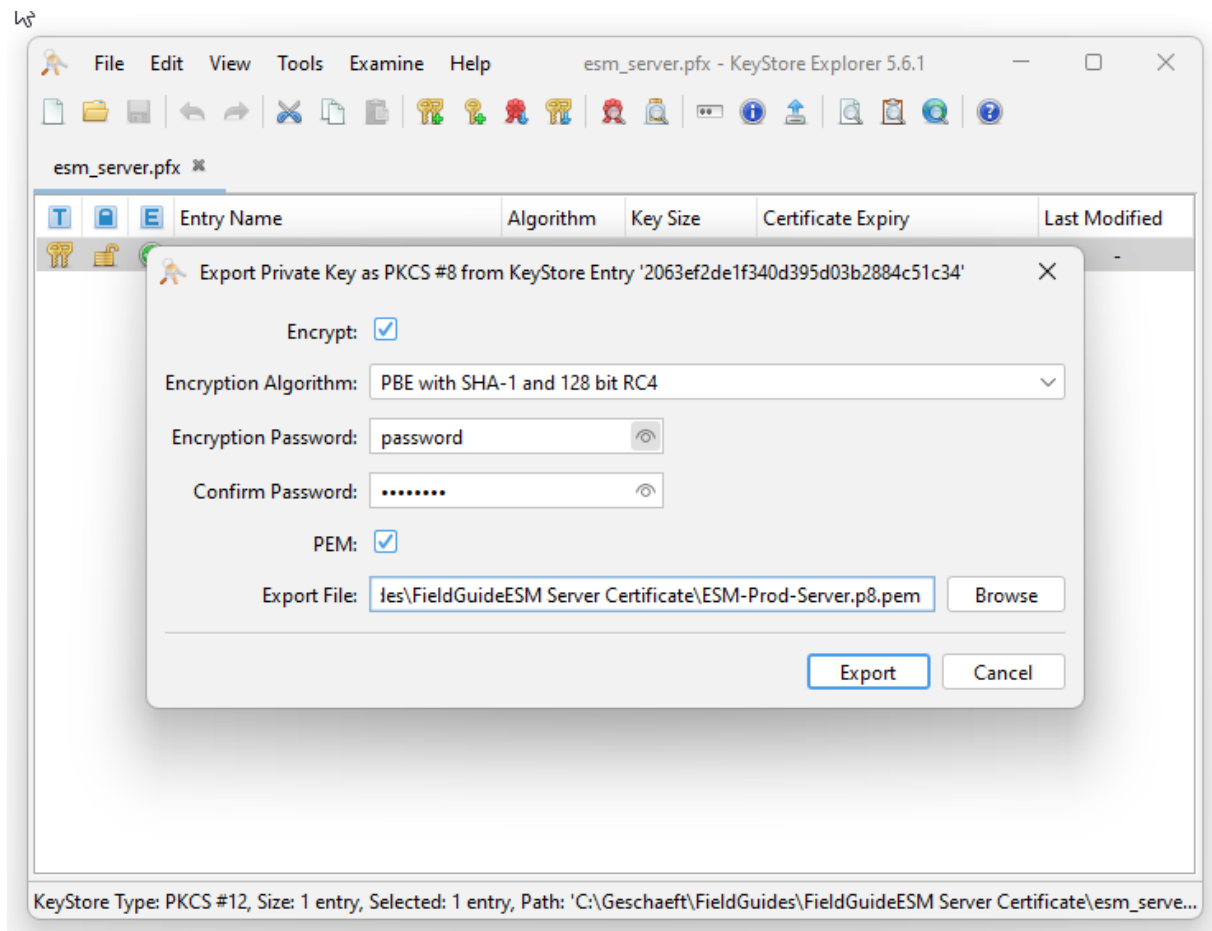
Right click on the certificate again and select Export -> Export Private Key:



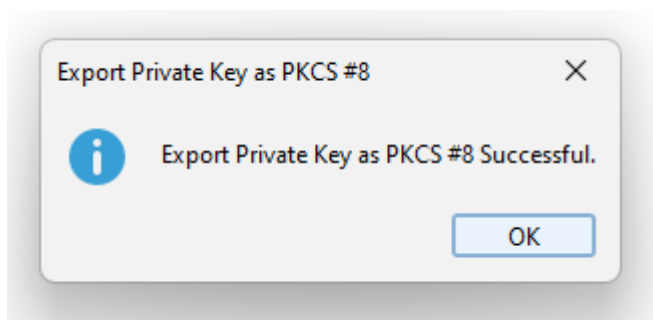
Keep the setting PKCS #8 and click OK.



Keep the settings as they are, for the password please use the word password – as this is used within the ESM Server. Also, you might want to rename the file to something else – e.g. ESM-Prod-Server.p8.pem and export the file to your destination folder:



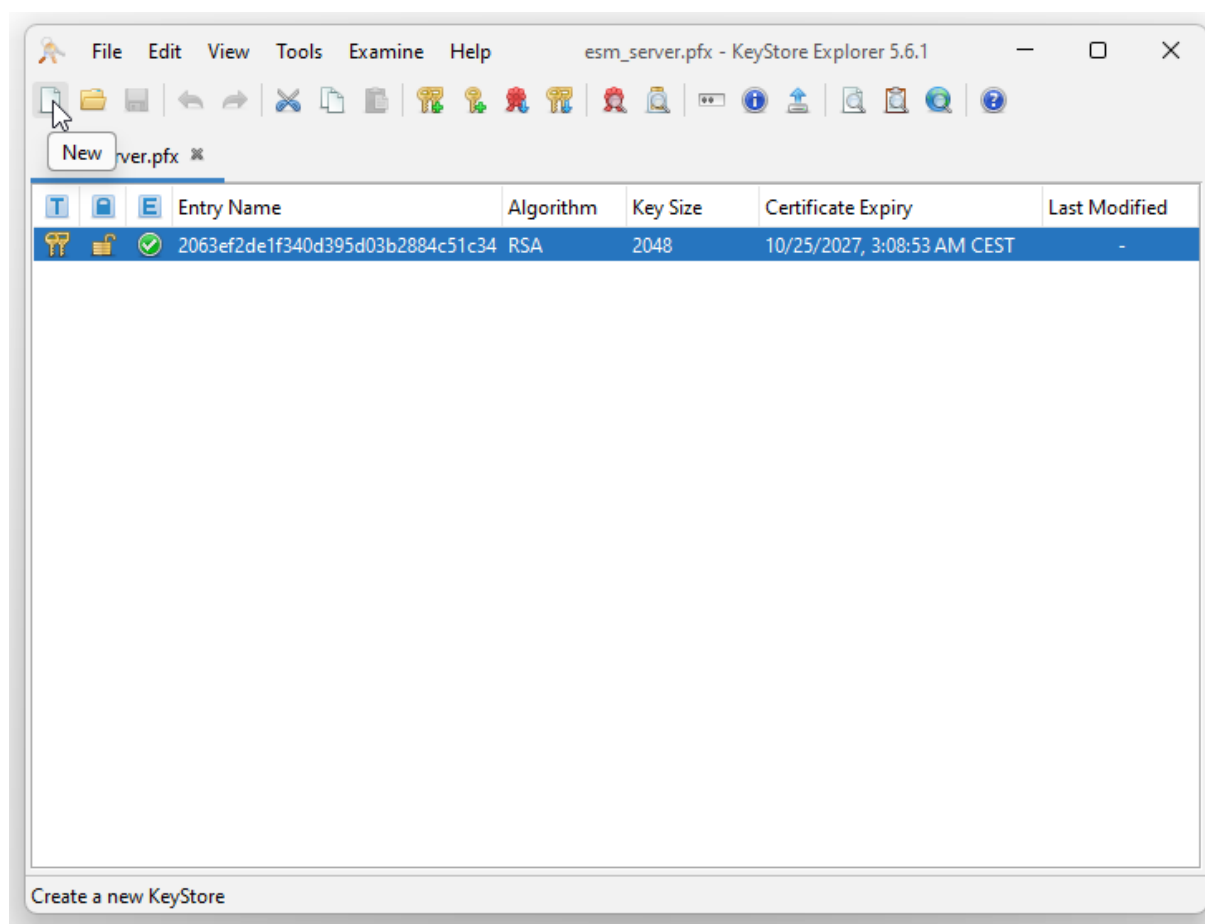
Click OK



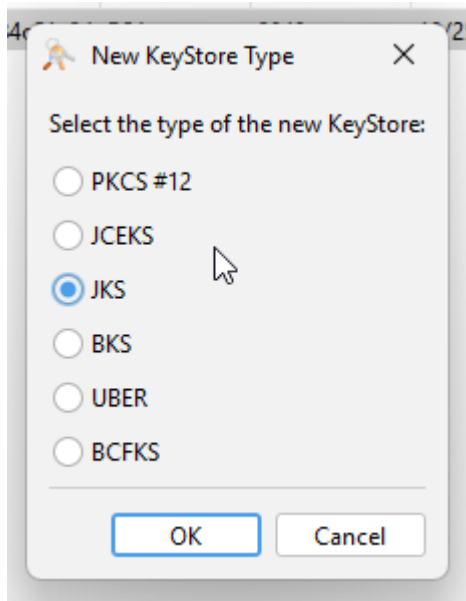
The export step is now completed, proceed with the next step.

3. Import the exported files in the suitable keystore for ESM Server

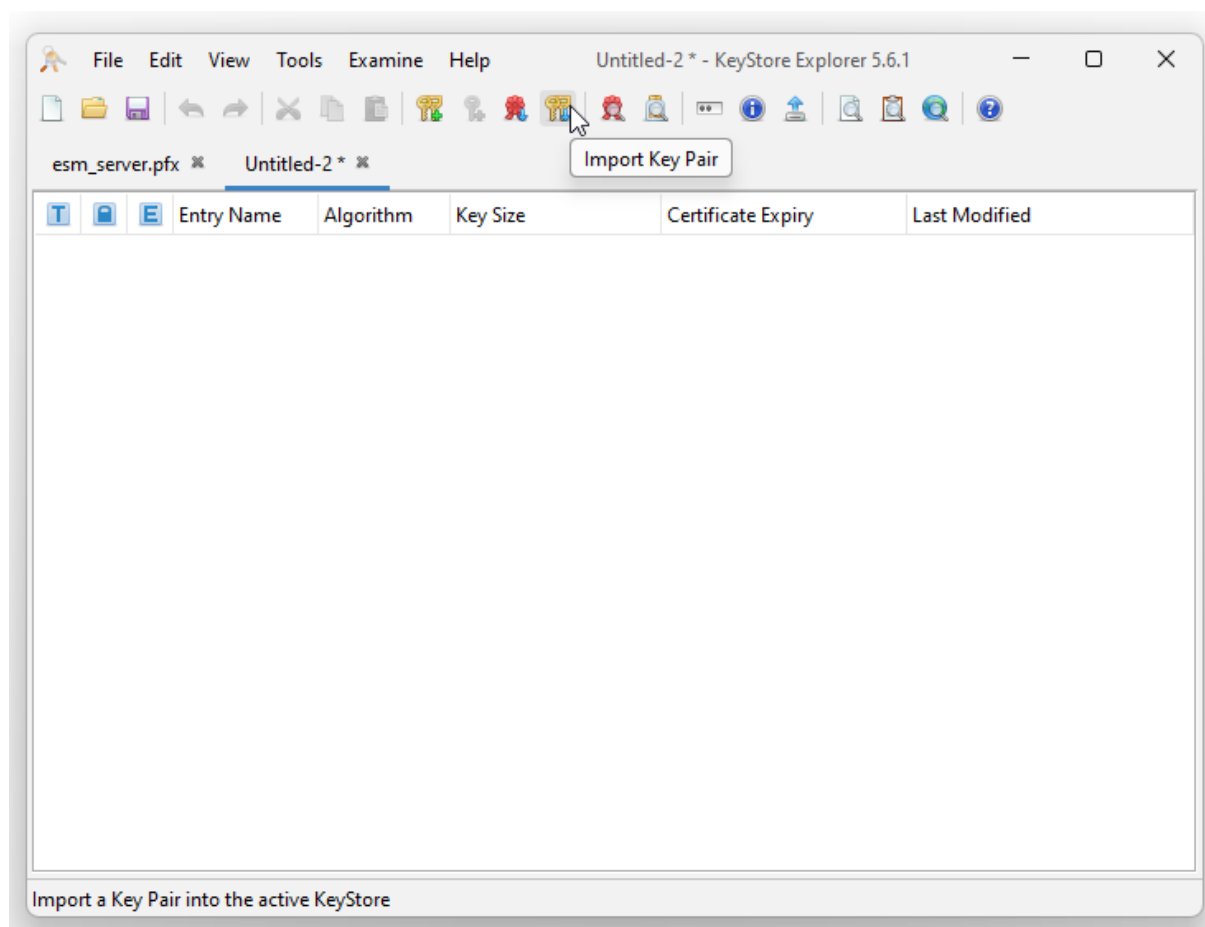
In the KeyStore Explorer tool click on “Create a new KeyStore”:



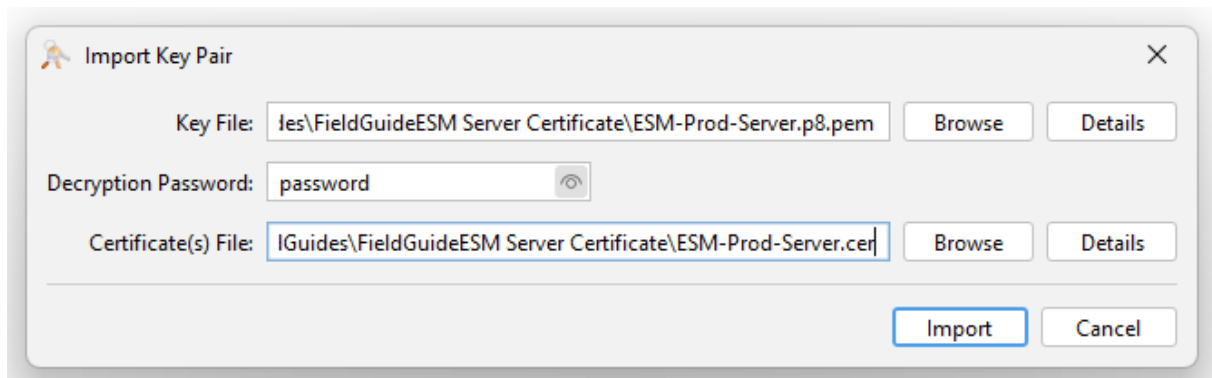
Select JKS and click OK:



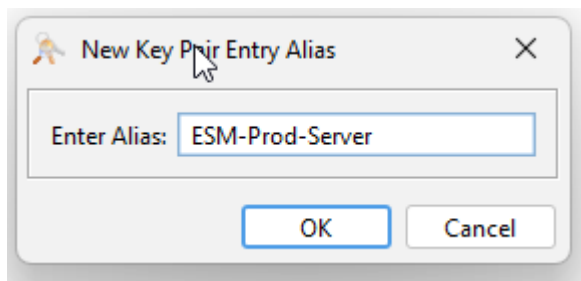
Click on “Import Key Pair”:



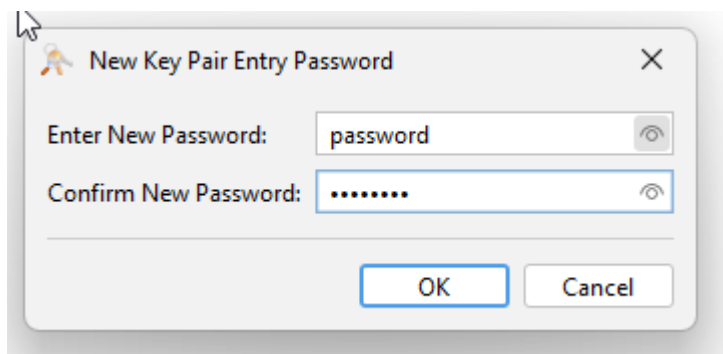
Browse your exported Key File, use the word password for the decryption password and browse the Certificate File – Click on Import to import both files in the new keystore:



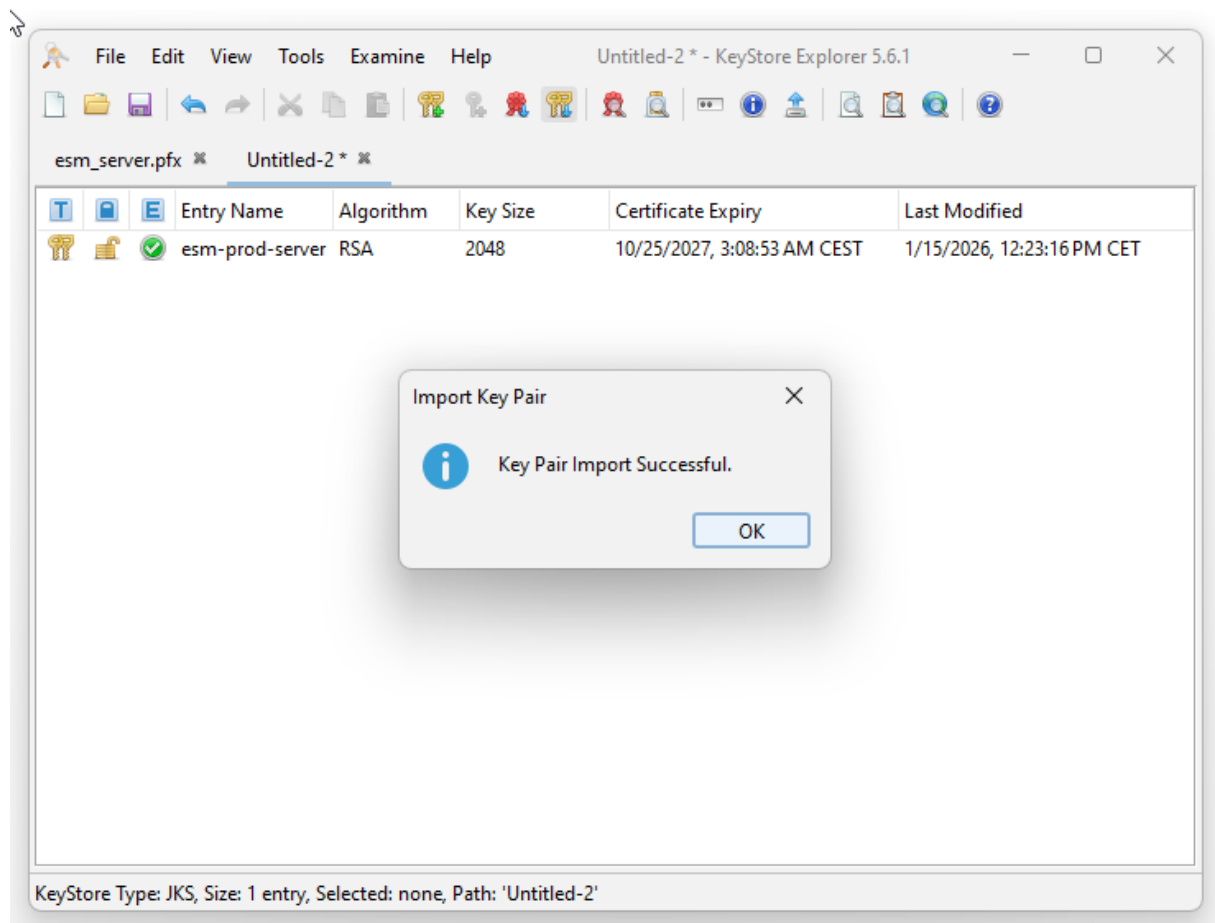
You will see an alias which you might want to change – Click on OK:



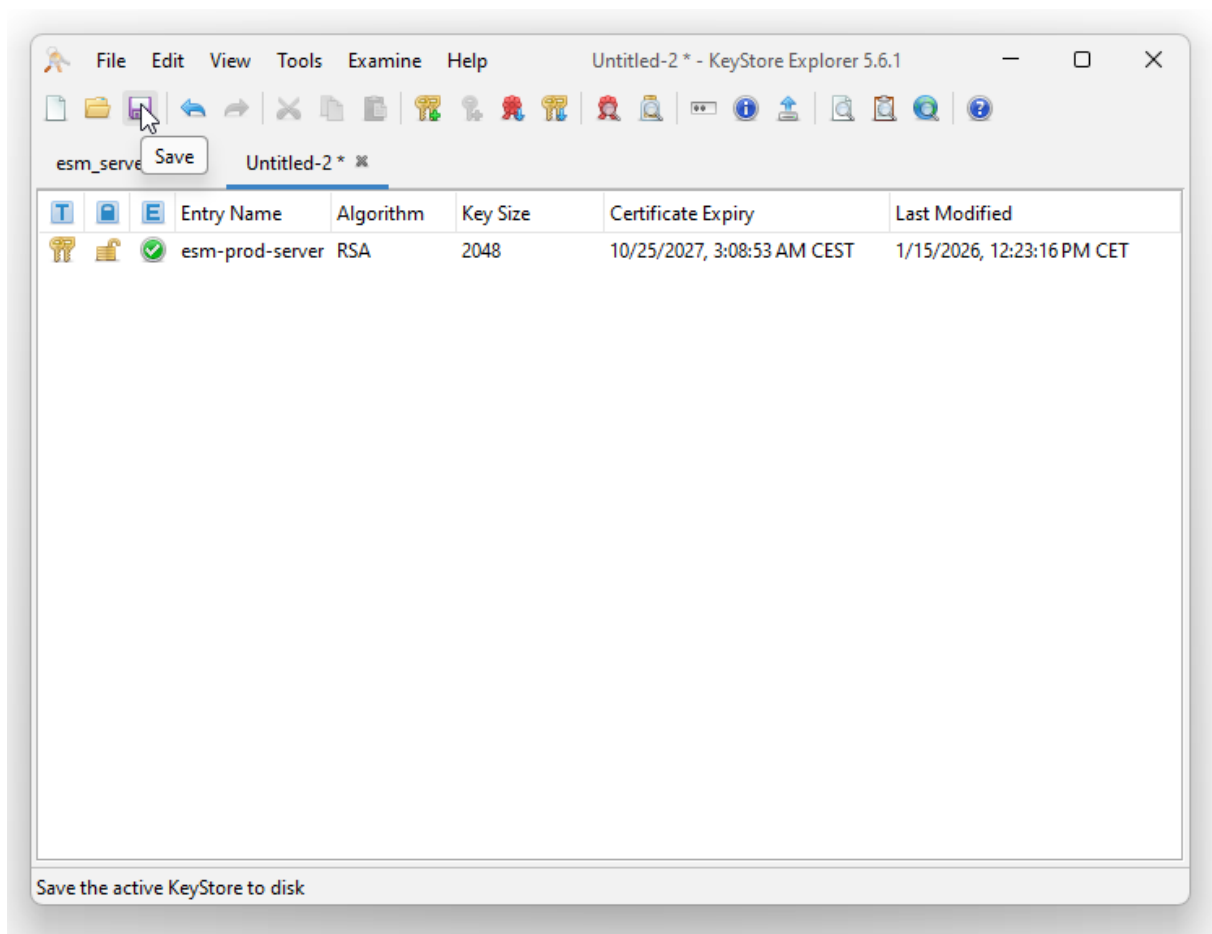
Specify the word password as Key Pair Entry Password and click OK:



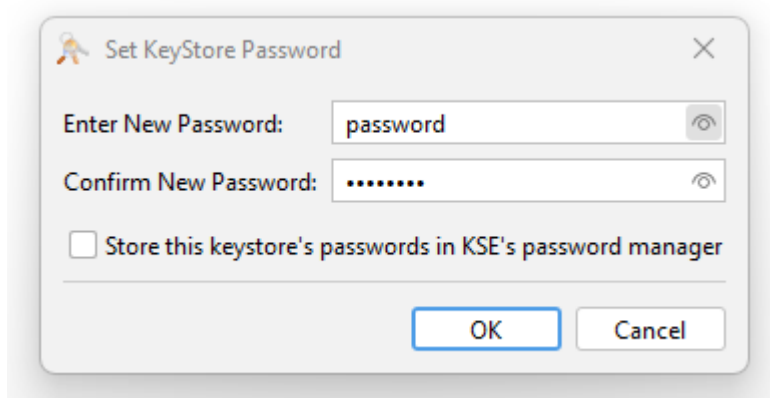
Click OK:



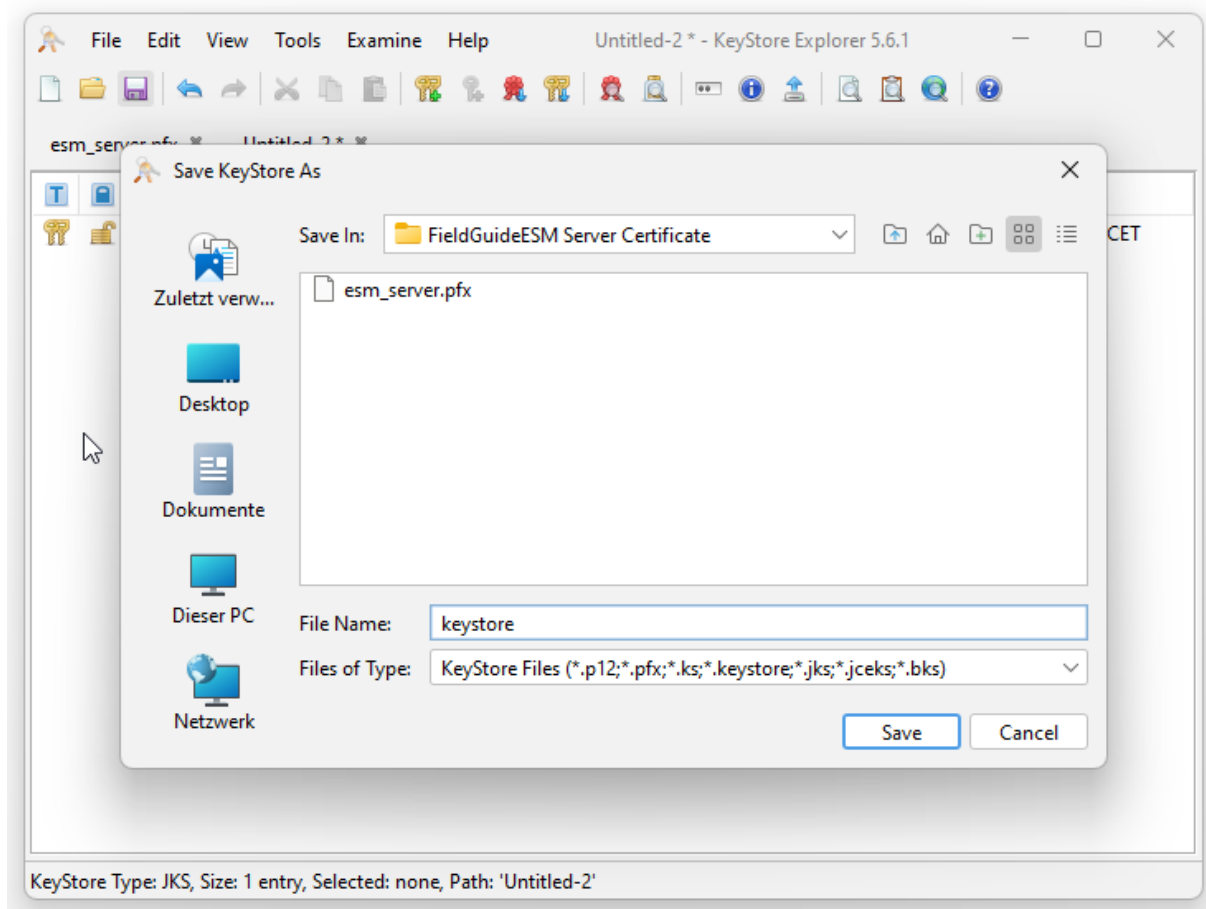
Save the newly created keystore – Click on the save Icon:



Again, specify the word password as KeyStore Password and click OK:



Select your destination folder and use “keystore” for the File Name – do not add any extension to the filename itself – Click on Save :

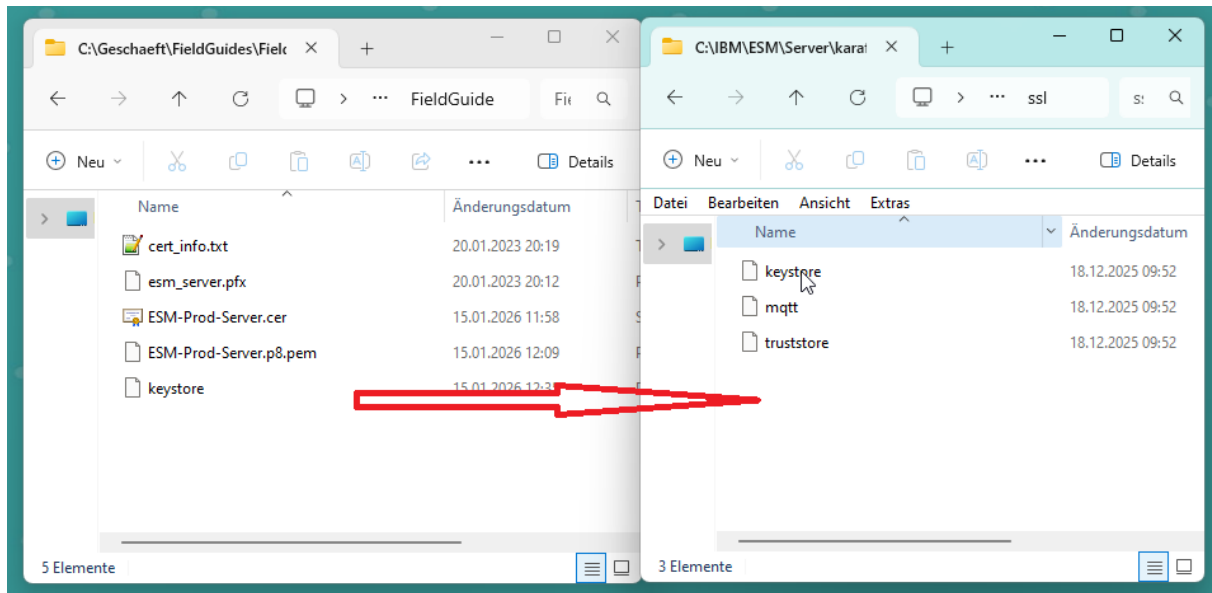


You can close the tool now if you want.

4. Exchange the keystore file on the ESM Server

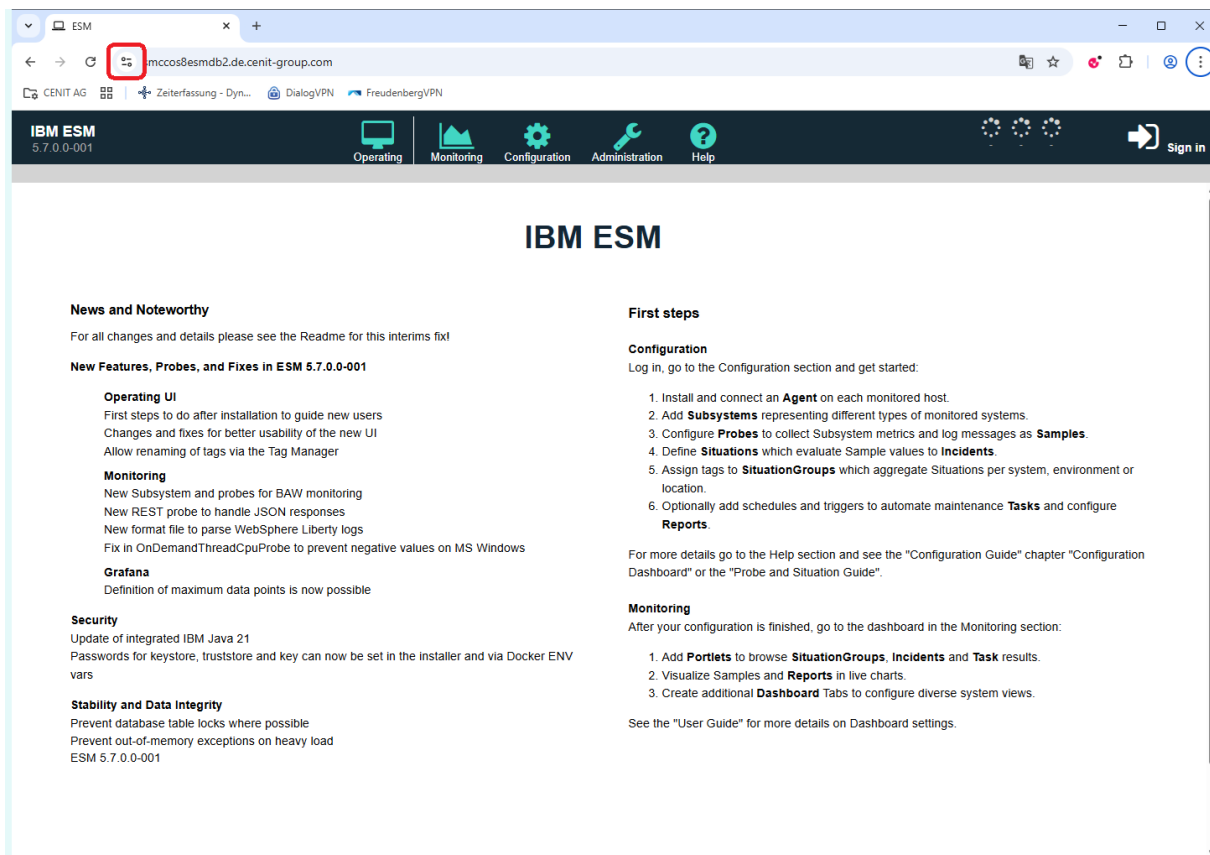
Stop the ESM Server Software.

Locate the keystore file you just created and replace the file in the Installation folder of ESM Server <Install-Dir>/karaf/etc/ssl with the new file (create a backup first):



Start ESM Server Software.

The browser should now show a valid Certificate for your console:



Contact Information

If you have any questions, please contact us at ECM.SystemMonitor@cenit.com.

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